This Zenodo deposit contains a publicly available description of the Dataset:

"Single nuclei ATAC sequencing of postmortem cingulate cortex,midbrain and motor cortex of healthy donors and Parkinson's disease patients – Scale-ATAC + 10x Genomics.".

Dataset Description:

This dataset consists of raw sequencing ATAC-seq data (Scale-ATAC pre-indexing followed by 10x Genomics snATAC v2). The data is part of an overall set of samples derived from postmortem midbrain (n=140), cingulate cortex (n=190) and motor cortex (n=4) of healthy donors (n=114), patients with Parkinson's disease (n=75) or patients with other neurological disorder (n=1). The protocol followed to isolate nuclei from postmortem brain samples and to prepare sequencing libraries can be found here: https://dx.doi.org/10.17504/protocols.io.14egnrb7ql5d/v1, https://dx.doi.org/10.17504/protocols.io.4r3l2z24ql1y/v1, https://cdn.10xgenomics.com/image/upload/v1666737555/support-documents/CG000496_Chromium_NextGEM_SingleCell_ATAC_ReagentKits_v2_UserGuide_RevB.pdf To increase throughput and to decrease batch effects, several donors have been pooled together into a single sequencing library. To computationally demultiplex the nuclei to their corresponding donors, cellsnp-lite (version commit: aad18644adcde853c313362a856a24245c9b91f7) followed by vireo (https://github.com/single-cell-genetics/vireo/pull/108) has been used. The population VCF with the donor genotypes derived from whole genome sequencing data has been used to assign nuclei back to their donors.

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ASAP Team: Voet

Dataset Name: voet-pmdbs-sn-atacseq-scalebio-10x, v0.1

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Publication DOI: NA

Grant IDs: ['ASAP-000430']

ASAP Lab: Aerts Lab

ASAP Project: Understanding inherited and acquired genetic variation in Parkinson's disease through single-cell multi-omics analyses: a unique data resource

Project Description: The functional roles of inherited and acquired genetic variation in the pathogenesis of Parkinson's disease (PD) remain largely unknown. Here, we will first study how germline genetic variants at PD-risk loci, which were identified by genome-wide association study (GWAS), perturb the expression of genes in specific cell (sub)populations of the brain and gut. To this aim, we will apply single-cell gene-expression and open-chromatin quantitative trait locus (QTL) analyses, enabling identification of PD-relevant genes and cell (sub)types. We will deliver the mechanisms of PD-candidate gene expression (dys)regulation in the normal condition, with ageing and in PD, as well as a unique single-cell multi-omic resource for the community. Second, we will study the nature and role of somatic mutations in brain and gut cells in PD- etiopathology. Finally, we will characterize biochemical and phenotypic effects of loss- or gain-of-function QTLs and somatic mutations of candidate genes in in vitro and in vivo model systems, including their impact on the neuro-immune axis.

Submission Date: 2026-03-12

This dataset is made available to researchers via the ASAP CRN Cloud: cloud.parkinsonsroadmap.org. Instructions for how to request access can be found in the User Manual.

This research was funded by the Aligning Science Across Parkinson's Collaborative Research Network (ASAP CRN), through the Michael J. Fox Foundation for Parkinson's Research (MJFF).